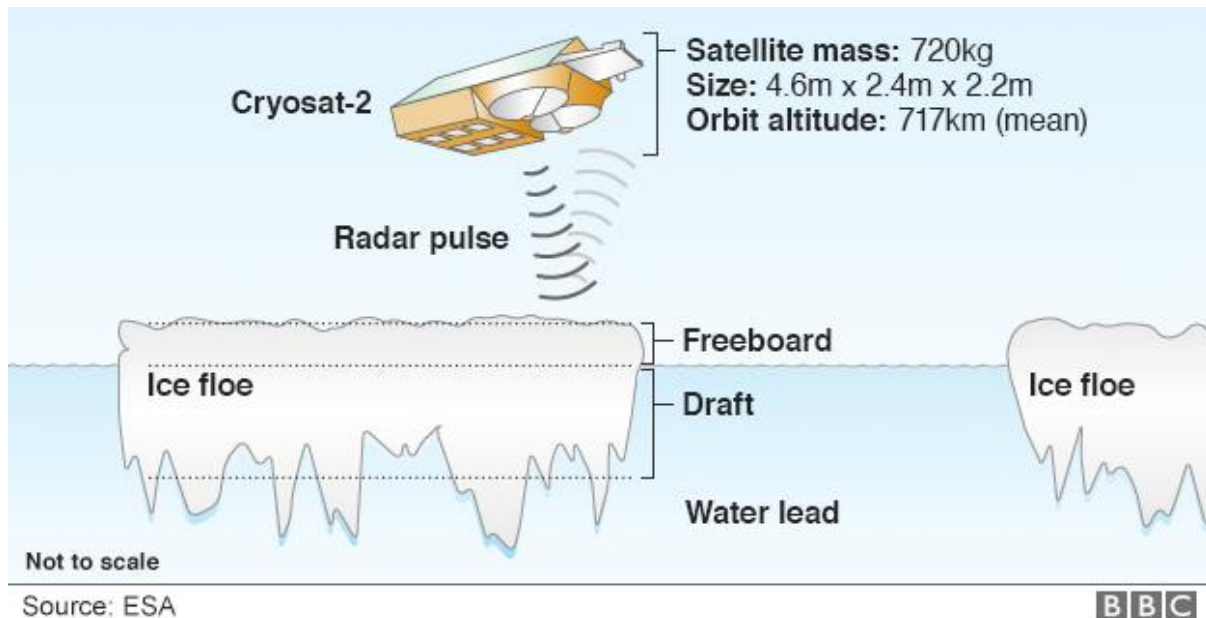


Tutorial-04 Solution

Cryosat-2 problem specification

Given:



The radar senses the height of the freeboard - the ice above the waterline – it is the top $\frac{1}{9}$ th of the floe. Assume the freeboard is a flat surface (no peaks or troughs etc).

Knowing this $\frac{1}{9}$ th figure allows Cryosat to work out sea-ice thickness

Some $\frac{8}{9}$ ^{ths} of the ice tends to sit below the waterline - the draft

The thickness multiplied by the area of ice cover produces a volume

Output the freeboard, the draft, the area, the thickness, the volume

- 0. initialise
 - 0.1 set freeboard
 - 0.2 set draft
 - 0.3 set volume
 - 0.4 set thickness
 - 0.5 set length
 - 0.6 set breadth

- 1. input
 - 1.1 input freeboard
 - 1.2 input length
 - 1.3 input breadth

- 2. process
 - 2.1 calc thickness = freeboard * 9
 - 2.2 calc draft = thickness - freeboard
 - 2.2 calc area = length * breadth
 - 2.3 calc volume = thickness * area

- 3. output
 - 3.1 print freeboard
 - 3.2 print draft
 - 3.3 print area
 - 3.4 print thickness
 - 3.5 print volume

Some variation between individual designs is normal, e.g. optimising the calculations. However, this is a reference point for programming so optimisation should not be done to the point of losing clarity, i.e. the calculation etc is too hard or complex to understand.

All variants of the pseudocode should all address the same issue - indicate the flow of logic required to meet the specification and identify key inputs.

Notes:

There is no recognised national standard – ‘house style’ prevails.

A ‘good’ solution should:

a) Should not exceed two sides of A4 paper

b) Should be consistent in the use of the sub-set of the English language.
See notes for guidelines for terms to use for our ‘house style’

c) Should be recognised that there is no ‘model answer’.
Variations are okay, e.g. the form of the algorithm for the calculations etc.
However, this is a communication document – don’t make calculations too complex so only you can understand it.

d) Should identify and clarify the logic, key inputs and major processing

e) Should always be language-independent.
It can be coded in any language so no specific programming-language terms should be included